A unique upper transition attractor for bounded similarity families

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Abstract

We prove Vince's Conjecture 8.1 for every bounded one-parameter family in the source's sense, in every finite dimension. The powers of the distinguished isometry need not be periodic. Damped powers converge as a compact set of operators, and regrouping the original maps gives a contraction constant bounded strictly below one throughout the limiting process.

Task 1. Prove uniqueness of the upper transition attractor. Here "bounded" has the technical meaning of Definition 8.1 in [1]: the family consists of similarities, each map has a fixed point independent of the parameter, and exactly one underlying similarity has the largest scaling ratio. It is not an assumption merely about bounded sets for an arbitrary affine family.

By rescaling the parameter so that the threshold is 1 and translating the special fixed point to 0, the source's Proposition 5.4 gives the form

$$\mathcal{F}_t = \{g_t, f_{1,t}, \dots, f_{m,t}\}, \quad g_t(x) = tUx, \quad f_{i,t}(x) = tL_i(x - q_i) + q_i, \tag{1}$$

where $U \in O(d)$, $q_i \in \mathbb{R}^d$, and $\rho := \max_i \|L_i\| < 1$. For the original problem each L_i is a similarity; the proof below only uses its displayed norm bound. Let A_t be the unique attractor for $0 \le t < 1$, with distances between compact sets measured by the Hausdorff metric h.

Theorem 1. *For the family* (1)*, A_t has a limit A_* as $t \uparrow 1$. Consequently it has a unique upper transition attractor, and the map $t \mapsto A_t$ is uniformly continuous on* $(0, 1)$*.*

We first isolate the compactness step. Operator norms are Euclidean norms, and h_{op} is the resulting Hausdorff metric on compact sets of matrices.

Lemma 1. *For $U \in O(d)$ set*

$$H = \overline{\{U^n : n \ge 0\}}, \quad E_t = \{0\} \cup \{t^n U^n : n \ge 0\} \quad (0 < t < 1), \quad D = \{rQ : 0 \le r \le 1,\ Q \in H\}.$$

Then H is a compact group, E_t and D are compact, and $h_{\mathrm{op}}(E_t, D) \to 0$ as $t \uparrow 1$.

Proof. The set H is a closed subsemigroup of the compact group $O(d)$. Choose an increasing sequence n_k such that U^{n_k} converges. Then $U^{n_{k+1}-n_k} \to I$, so $U^{n_{k+1}-n_k-1} \to U^{-1}$, and $U^{-1} \in H$. Thus all integral powers belong to H, and continuity of inversion shows that H is a group. Compactness of D follows from its parametrization by $[0, 1] \times H$. For fixed $t < 1$, the only possible accumulation point of $t^n U^n$ as $n \to \infty$ is 0, so E_t is compact.

Fix $\varepsilon > 0$. Density of the nonnegative powers and compactness give an integer $L \ge 0$ such that $I, U, \dots, U^L$ is an ε-net of H. For every $N \ge 0$, the list $U^N, \dots, U^{N+L}$ is also an ε-net, since left multiplication by U^N is an isometry of H.

Given $rQ \in D$ with $r > 0$, choose $N = \lfloor \log r / \log t \rfloor$, so $t^{N+1} < r \le t^N$. Some $j \in \{0, \dots, L\}$ satisfies $\|U^{N+j} - Q\| \le \varepsilon$. It follows that

$$\begin{aligned} \|t^{N+j}U^{N+j} - rQ\| &\le \varepsilon + |t^{N+j} - r| \\ &\le \varepsilon + (1 - t^L) + (1 - t). \end{aligned}$$

The case $r = 0$ is exact, since $0 \in E_t$. Conversely $E_t \subseteq D$. Thus

$$h_{\mathrm{op}}(E_t, D) \le \varepsilon + (1 - t^L) + (1 - t). \tag{2}$$

First let $t \uparrow 1$ with ε, L fixed, then let ε decrease to zero. No periodicity assumption has been used. □

Proof of Theorem 1. Choose $C \ge \sup_{i,\, 0 \le t \le 1} \|f_{i,t}(0)\|$ and a radius $R > \max\{1, C/(1-\rho)\}$. The closed ball $B = \overline{B}(0, R)$ is invariant under all the maps in (1), including at $t = 1$:

$$\|f_{i,t}(x)\| \le \rho R + C \le R, \qquad \|g_t(x)\| \le R \quad (x \in B).$$

In particular $A_t \subseteq B$. Define compact families of maps on B by

$$\mathcal{M}_t = \{x \mapsto M f_{i,t}(x) : M \in E_t,\ 1 \le i \le m\}, \qquad \mathcal{M}_* = \{x \mapsto M f_{i,1}(x) : M \in D,\ 1 \le i \le m\}.$$

Each family includes the constant-zero map. Every map in these families has Lipschitz constant at most ρ, uniformly in t and M. Their Hutchinson operators Φ_t, Φ_* on the complete metric space $\mathcal{K}(B)$ of nonempty compact subsets of B are therefore ρ-contractions. Their values are compact because the parameter sets and the input sets are compact. The contraction theorem gives a unique fixed compact set for each operator; denote the fixed set of Φ_* by A_*.

We claim that the fixed set of Φ_t is exactly A_t. Put $F_t(K) = \bigcup_i f_{i,t}(K)$. Iterating the original invariance equation gives

$$A_t = \bigcup_{n=0}^{k-1} g_t^n(F_t(A_t)) \ \cup \ g_t^k(A_t) \qquad (k \ge 1). \tag{3}$$

Also $0 \in A_t$, since repeated application of g_t to any point of A_t converges to 0. Every set in the union on the right lies in A_t. Conversely, if a point of A_t lies in none of these sets for any n, then (3) places it in $g_t^k(A_t)$ for every k, so its norm is at most $t^k R$ for every k and it is zero. Since $g_t^n(x) = t^n U^n x$,

$$A_t = \{0\} \cup \bigcup_{n \ge 0,\, i} g_t^n(f_{i,t}(A_t)) = \Phi_t(A_t).$$

This proves the claim, including the required infinite tails.

Let

$$\eta_t = h_{\mathrm{op}}(E_t, D), \qquad e_t = \max_i \sup_{x \in B} \|f_{i,t}(x) - f_{i,1}(x)\|.$$

Lemma 1 gives $\eta_t \to 0$. Formula (1) gives $e_t \le (1-t) \max_i \sup_{x \in B} \|L_i(x - q_i)\| \to 0$. Pairing close matrices in the two coefficient sets, using $f_{i,1}(B) \subseteq B$ and $\|M\| \le 1$, shows that for every $K \in \mathcal{K}(B)$,

$$h(\Phi_t(K), \Phi_*(K)) \le e_t + R\eta_t.$$

The fixed-point equations now imply

$$h(A_t, A_*) \le \rho h(A_t, A_*) + e_t + R\eta_t, \qquad \boxed{h(A_t, A_*) \le \frac{e_t + R\eta_t}{1 - \rho} \ \longrightarrow \ 0.}$$

Hence the upper transition attractor is unique.

For completeness, on every interval $[0, b]$ with $b < 1$, the original Hutchinson operators have a common contraction factor at most b and depend uniformly continuously on t on B. The same fixed-point estimate proves continuity of A_t there, including at 0, where the attractor is the finite set of constant-map values. Together with the established limit at 1, this extends $t \mapsto A_t$ continuously to $[0, 1]$ and thus proves uniform continuity. Rescaling back gives the assertion on $(0, t_0)$. □

If the original family has only the distinguished map, its attractor is the special fixed point for every $t < 1$, so the conclusion holds immediately. The limiting set above need not be an attractor of the original threshold IFS $\mathcal{F}_1$; only uniqueness of the upper transition attractor is asserted.

The final 2022 article [2], p. 5418, explicitly describes its periodic-isometry theorem as a partial answer to Vince's conjecture. Its Question 6.2 on p. 5425 leaves the nonperiodic case open. The compact operator-set limit in Lemma 1 supplies that missing step for the finite-dimensional source problem.

Task 2. Quantify convergence for nonperiodic rotations. Use covering times of $\{U^n\}$ in H and (2) to derive rates for $h(A_t, A_*)$. Determine the sharp rates under Diophantine conditions on the rotation angles and contrast them with the linear bound available when U is periodic.

Task 3. Determine useful infinite-dimensional hypotheses. Extend the argument to Banach spaces under compactness assumptions on the isometry powers. Determine whether compactness of the relevant point orbits can replace compactness in operator norm.

Task 4. Treat several maps at the critical scaling ratio. Determine when multiple distinguished similarities admit a unique limiting attractor. Analyze the roles of a common fixed point, the group they generate, and boundedness of the attractors near the threshold.

Task 5. Analyze the geometry of the limiting set. Use the explicit family $\mathcal{M}_*$ to characterize the radial, symmetry and interior properties of A_*. Determine when this set agrees with the lower transition attractor and when the containment is strict.

References

[1] A. Vince, *Thresholds for one-parameter families of affine iterated function systems*, Nonlinearity **33** (2020), 6541–6563. Final journal PDF; arXiv:1912.03220. Definition 8.1 and Conjecture 8.1 appear on pp. 6557–6558.

[2] K. Leśniak, N. Snigireva, F. Strobin and A. Vince, *Transition phenomena for the attractor of an iterated function system*, Nonlinearity **35** (2022), 5396–5426. doi:10.1088/1361-6544/ac8af1. Theorem 5.2, p. 5418, and Question 6.2, p. 5425.